

INDRANEIL PAUL

 Email  Github  HuggingFace  Scholar  Twitter  LinkedIn  Website

I am a doctoral researcher interested in optimizing **code generation LM pre-training and post-training**, with an emphasis on **tool usage** and **multilingual performance**. My long-term mission is to enhance model tool use, unlocking the application of LMs beyond text-only settings to areas such as **robot navigation** and **agentic workflows** by improving their capabilities to **reason**, **offload computation**, and learn from **environmental feedback**. I also work on **preference learning** methods to enhance LMs' code generation capabilities along non-functional axes, such as **security** and **efficiency**. My interests span all facets of improving LM training efficacy, including **data curation**, **context length extension**, **modularity**, and **reinforcement learning**. I am also a long-standing open-source contributor to multiple public repositories and have worked on numerous **open-source LM** releases.

EDUCATION

09/22 – Pres.	ELLIS PhD Candidate in Informatics, TU Darmstadt, Germany	 ENROLMENT
07/17 – 07/19	Masters by Research in Computer Science, IIIT Hyderabad, India	 CERTIFICATE
08/13 – 05/17	Bachelors of Technology in Computer Science, IIIT Hyderabad, India	 CERTIFICATE

SELECTED PUBLICATIONS

THEMIS : TRAINING ROBUST MULTILINGUAL CODE REWARD MODELS FOR FLEXIBLE MULTI-CRITERIA SCORING TMLR 2026 (Under Review) Indraneil Paul et al.	 ABSTRACT  PDF
ALETHEIA : WHAT MAKES RLVR FOR CODE VERIFIERS TICK? EMNLP 2026, Budapest (Under Review) Vatsal Venkatkrishna et al. (incl. Indraneil Paul)	 ABSTRACT  PDF
AICD BENCH : A CHALLENGING BENCHMARK FOR AI-GENERATED CODE DETECTION EACL 2026, Rabat Daniil Orel et al. (incl. Indraneil Paul)	 ABSTRACT  PDF
DROID : A RESOURCE SUITE FOR AI-GENERATED CODE DETECTION EMNLP 2025, Suzhou Daniil Orel et al. (incl. Indraneil Paul)	 SLIDES  ABSTRACT  PDF
OBSCURACODER : POWERING EFFICIENT CODE LM PRE-TRAINING VIA OBFUSCATION GROUNDING ICLR 2025 Poster, Singapore Indraneil Paul et al.	 SLIDES  ABSTRACT  PDF
BIGCODEBENCH : BENCHMARKING CODE GENERATION WITH DIVERSE FUNCTION CALLS AND COMPLEX INSTRUCTIONS ICLR 2025 Oral, Singapore Terry Yue Zhuo et al. (incl. Indraneil Paul)	 SLIDES  ABSTRACT  PDF
IRCORDER : INTERMEDIATE REPRESENTATIONS MAKE LANGUAGE MODELS ROBUST MULTILINGUAL CODE GENERATORS ACL 2024 Oral, Bangkok ( Outstanding Paper) Indraneil Paul et al.	 SLIDES  ABSTRACT  PDF
STARCORDER 2 AND THE STACK V2 : THE NEXT GENERATION TMLR 2024 Anton Lozhkov et al. (incl. Indraneil Paul)	 SLIDES  ABSTRACT  PDF
ADAPTERS : A UNIFIED LIBRARY FOR PARAMETER-EFFICIENT AND MODULAR TRANSFER LEARNING EMNLP 2023 System Demonstrations, Singapore Clifton Poth et al. (incl. Indraneil Paul)	 DEMO  ABSTRACT  PDF
SUB-TASK IMPUTATION VIA SELF-LABELLING TO TRAIN IMAGE MODERATION MODELS ON SPARSE NOISY DATA CIKM 2022 Oral, Atlanta Indraneil Paul et al.	 SLIDES  ABSTRACT  PDF

INDUSTRY EXPERIENCE

- 10/25 – 04/26 **Applied Scientist PhD Intern, Amazon Inc. (Web Services), Berlin**
- Researched RL methods to reduce the cloud tool-calling error-rate in **Amazon Q Developer** agent
 - Explored asynchronous RL approaches for improved distributed training efficiency
 - Automated RL environment creation for agents using infrastructure-as-code emulators
- HuggingFace Transformers** **VeRL** **Megatron-LM** **LocalStack** **AWS Fargate** **AWS AgentCore** **AWS SageMaker**
- 04/20 – 08/22 **Applied Scientist, Amazon Inc. (Advertising), Bangalore**
- Created text, image and multi-modal models for improving EU ad moderation automation by 28%
 - Researched multi-modal, multi-lingual and multi-task pre-training objectives for ad catalog tagging
 - Devised sample-efficient training methods for ViT models using self-labelling and sub-task distillation
- HuggingFace Transformers** **PyTorch** **CUDA C++** **TensorRT** **AWS SageMaker**
- 07/19 – 03/20 **Software Development Engineer, Amazon Inc. (Logistics), Hyderabad**
- Implemented a planner enabling merchants to rank options and schedule last-mile package drop-offs
 - Oversaw database tuning, JVM optimizations and message queue setup for event ingestion service
- Spring** **METIS** **Java** **AWS SNS** **AWS SQS** **AWS DynamoDB**

OPEN SOURCE EXPERIENCE

- 04/24 – 09/24 **MaLA-LM, UTTER Project**
- Conducted SOTA multilingual continual pre-training evaluations on frontier LMs
 - Investigated the code completion performance of multilingual LMs in non-English language prompts
 - Worked on the **EMMA-500** model and **MaLA-2** massively multilingual corpus releases
- HuggingFace Transformers** **Megatron-DeepSpeed** **DeepSpeed** **Docker**
- 06/23 – 09/24 **BigCode Project, ServiceNow and HuggingFace**
- Contributed to **StarCoder-2** pre-training data collection and training ablations
 - Worked on containerization, evaluation framework and annotation for **BigCodeBench**
- LLVM** **HuggingFace Transformers** **Megatron-LM** **Docker**
- 05/17 – 07/17 **Google Summer of Code, Green Navigation**
- Implemented an LSTM forecaster for the **EV-Charge-Prediction** project to alleviate range anxiety
 - Implemented an ensemble solution that reduced absolute forecasting error by 39%
 - Productionized the Bayesian Optimization service for optimal hyper-param selection in training jobs
- TensorFlow** **Pandas** **BayesOpt**

ACADEMIC SERVICE

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|-------|---|--|
| 07/26 | Organizer, SemEval Task on GenAI Code Detection & Attribution |  EVENT PAGE |
| 08/22 | Program Committee, KDD Content Generation for e-Commerce Workshop |  EVENT PAGE |

INVITED TALKS

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| 10/24 | Challenges in Code LMs, IIIT Hyderabad |  SLIDES |
| 09/24 | Code Generation : Challenges and Solutions, BHT Berlin |  SLIDES |
| 04/23 | Parameter-Efficient Fine-Tuning for NLP, MBZUAI |  SLIDES |
| 01/23 | Multilingual Adapters, TU Darmstadt |  SLIDES |

SUMMER SCHOOLS

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|-------|--|---|
| 07/23 | Lisbon Machine Learning Summer School (LxMLS) |  CERTIFICATE |
| 07/21 | European Summer School in Logic, Language and Information (ESSLLI) |  CERTIFICATE |

REFERENCES

- | | | |
|--------------|--|---|
| AWS Berlin | Dr. György Szarvas, Research Internship Advisor |  EMAIL |
| TU Darmstadt | Prof. Dr. Iryna Gurevych, PhD Thesis Advisor |  EMAIL |
| JMU Würzburg | Prof. Dr. Goran Glavas, PhD Thesis Co-Advisor |  EMAIL |
| IIIT Hyd. | Prof. Dr. Ponnurangam Kumaraguru, MSc Thesis Advisor |  EMAIL |